

Dietary patterns, food groups, and telomere length in the Multi-Ethnic Study of Atherosclerosis (MESA)



Background Telomere length reflects biological aging and may be influenced by environmental factors, including those that affect inflammatory processes. **Objective** With data from 840 white, black, and Hispanic adults from the Multi-Ethnic Study of Atherosclerosis, we studied cross-sectional associations between telomere length and dietary patterns and foods and beverages that were associated with markers of inflammation. **Design** Leukocyte telomere length was measured by quantitative polymerase chain reaction. Length was calculated as the amount of telomeric DNA (T) divided by the amount of a single-copy control DNA (S) (T/S ratio). Intake of whole grains, fruit and vegetables, low-fat dairy, nuts or seeds, nonfried fish, coffee, refined grains, fried foods, red meat, processed meat, and sugar-sweetened soda were computed with responses to a 120-item food-frequency questionnaire completed at baseline. Scores on 2 previously defined empirical dietary patterns were also computed for each participant. **Results** After adjustment for age, other demographics, lifestyle factors, and intakes of other foods or beverages, only processed meat intake was associated with telomere length. For every 1 serving/d greater intake of processed meat, the T/S ratio was 0.07 smaller ($\beta \pm \text{SE}$: -0.07 ± 0.03 , $P = 0.006$). Categorical analysis showed that participants consuming ≥ 1 serving of processed meat each week had 0.017 smaller T/S ratios than did nonconsumers. Other foods or beverages and the 2 dietary patterns were not associated with telomere length. **Conclusions** Processed meat intake showed an expected inverse association with telomere length, but other diet features did not show their expected associations.